PRIVACY-PRESERVING SHOPPING INTELLIGENCE

The project in numbers

What was built, what earlier experiments showed, why we changed course, and what is measured so far

Ahmed Abdelhamed · MSc, Queen's University · supervisor meeting, 30 September 2026 · facts as of 29 Sep, 18:36 UTC

AT A GLANCE

Eight numbers that summarise the work

158,486

products ranked for every prediction

1.45 M

users in the study pool (5–100 events each)

131,072

users in the main cohort

2.59 M

training decisions in the main cohort

4

phases: next category → next product (cloud) → features → deployment study

8

runs in the backbone contest (2 families × 2 sizes × 2 seeds)

+22.7%

federated gain so far from tuning (calibration users)

7 / 10

independent 14-part review; 8–8.5 after the fixes

Source: Data protocol and population lock; contest result; ladder register; review synthesis

THE JOURNEY

Four phases, each answering one question

1

Next-category MVP

Can we predict the next category?

0.3474

GRU vs 0.3143 transition table (Oct 2019)

2

Next product

How good can a cloud model get?

0.1713

best central model (SASRec), validation, full data

3

Features

Which columns help?

58 → 24

input columns; 21 of 24 computable on device

4

Deployment study

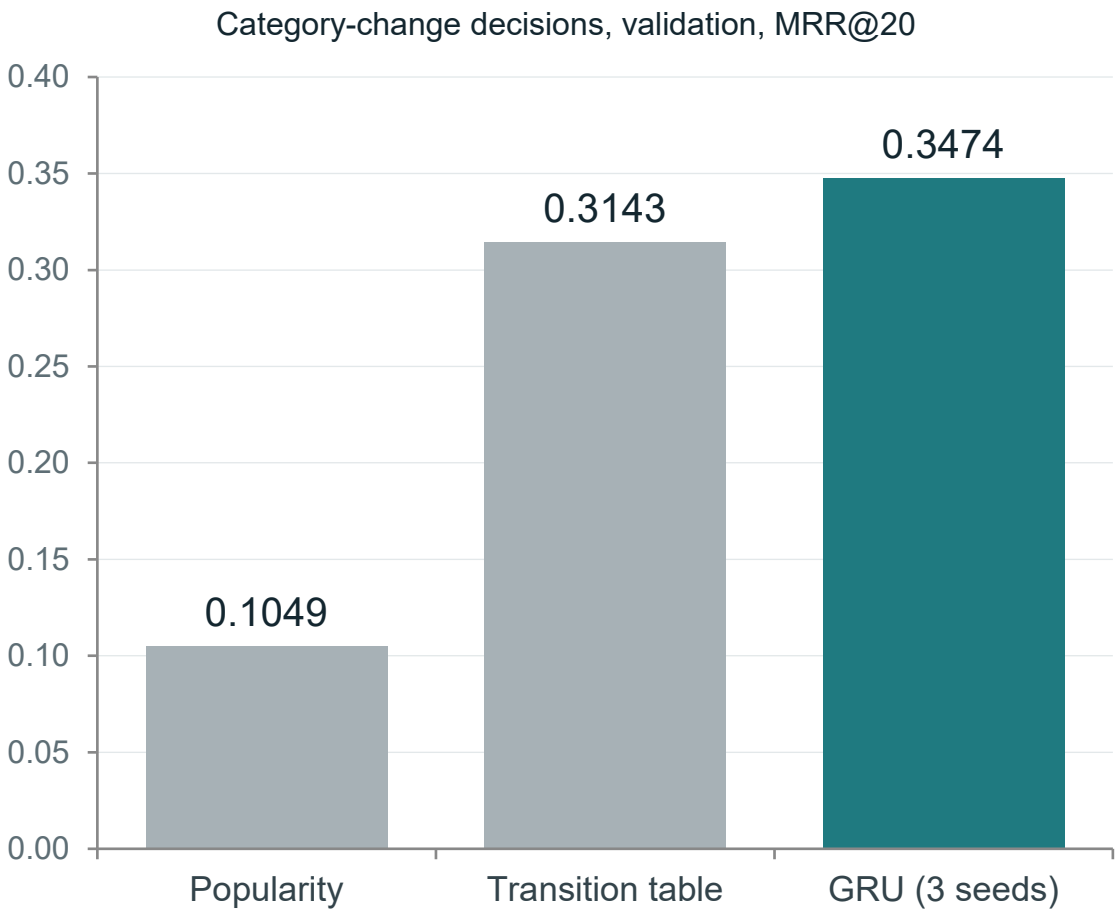
How much survives off the cloud?

+22.7%

federated tuning so far; main results sealed

Source: MVP report (next category); central-model campaign reports; feature dictionary and feature search; ladder register

Our next-category numbers



+0.0331

GRU over the transition table (seed spread 0.0007)

88.4%

federated pilot kept of centralised quality (1,000 clients)

82%

of the overall metric solved by a no-learning rule

+0.066

purchase PR-AUC gain: 0.076 (rule) → 0.142 (model)

Stronger or weaker than published work? Unknown — no published next-category results on REES46 to compare with.

Source: GitHub repo: docs/MVP_REPORT_AR.md §§1–2; ADR-001 (evaluation protocol); MVP federated pilot evidence (mvp-t1-003)

WHY WE MOVED TO NEXT-PRODUCT

Three reasons, in numbers

0

published next-category results on REES46 — no way to know if we are stronger or weaker

82%

of the next-category metric solved without learning — the task is artificially easy

1

official published next-product benchmark on REES46 (GRU4Rec, RecSys 2023) — a reference point

Next category **588 classes**

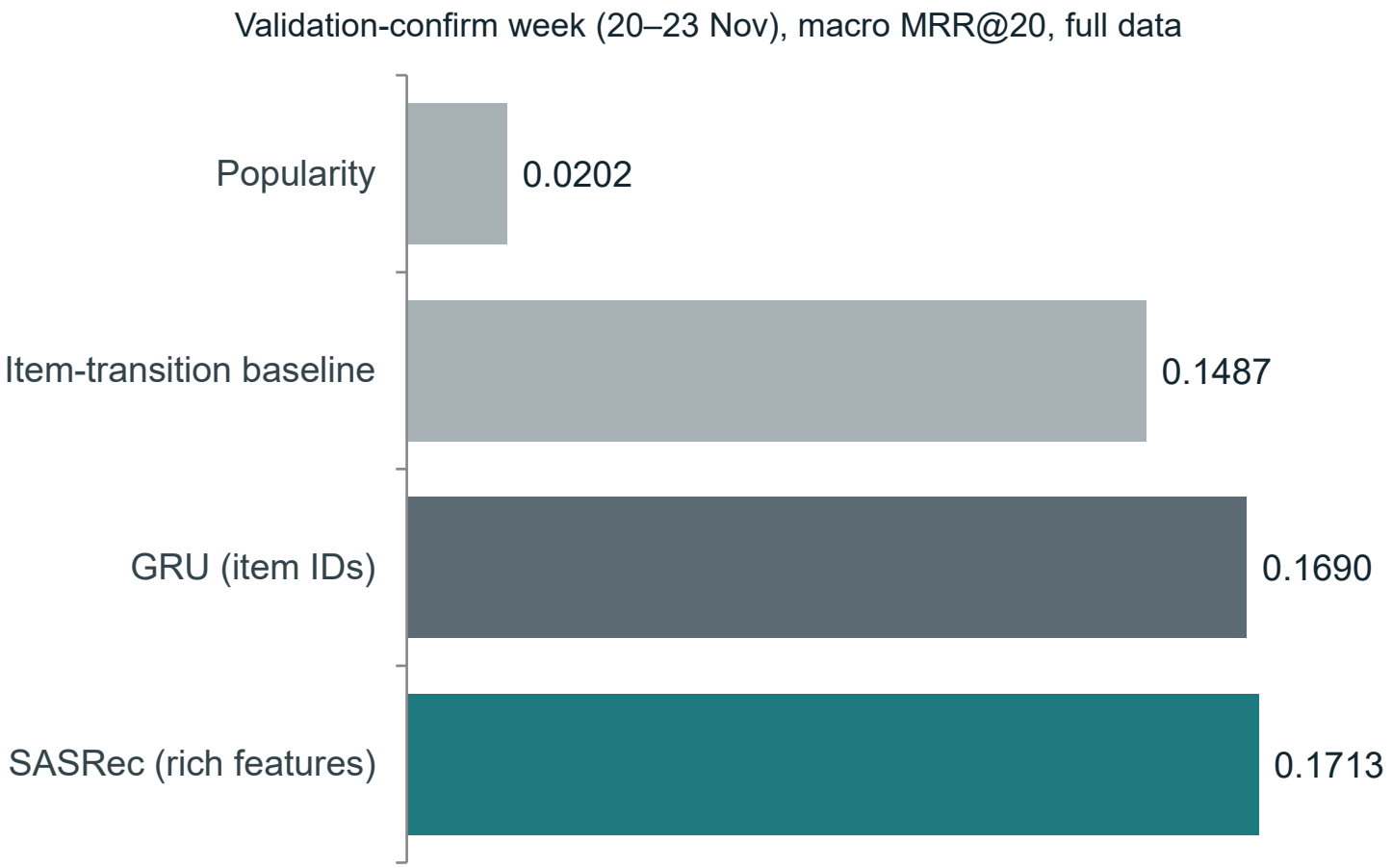
Next product **158,486 products**

Harder, more standard task — agreed with the supervisor.

Source: ADR-001; thesis evidence note (proposal amendment table); literature register card R05; thesis research review A3

EARLIER PHASE · full data · not comparable with the current study

On the full data, sequence models beat the baselines



52.8 M

training decisions (40.2M loss examples)

+0.0024

SASRec over GRU — but ID-only SASRec was below GRU

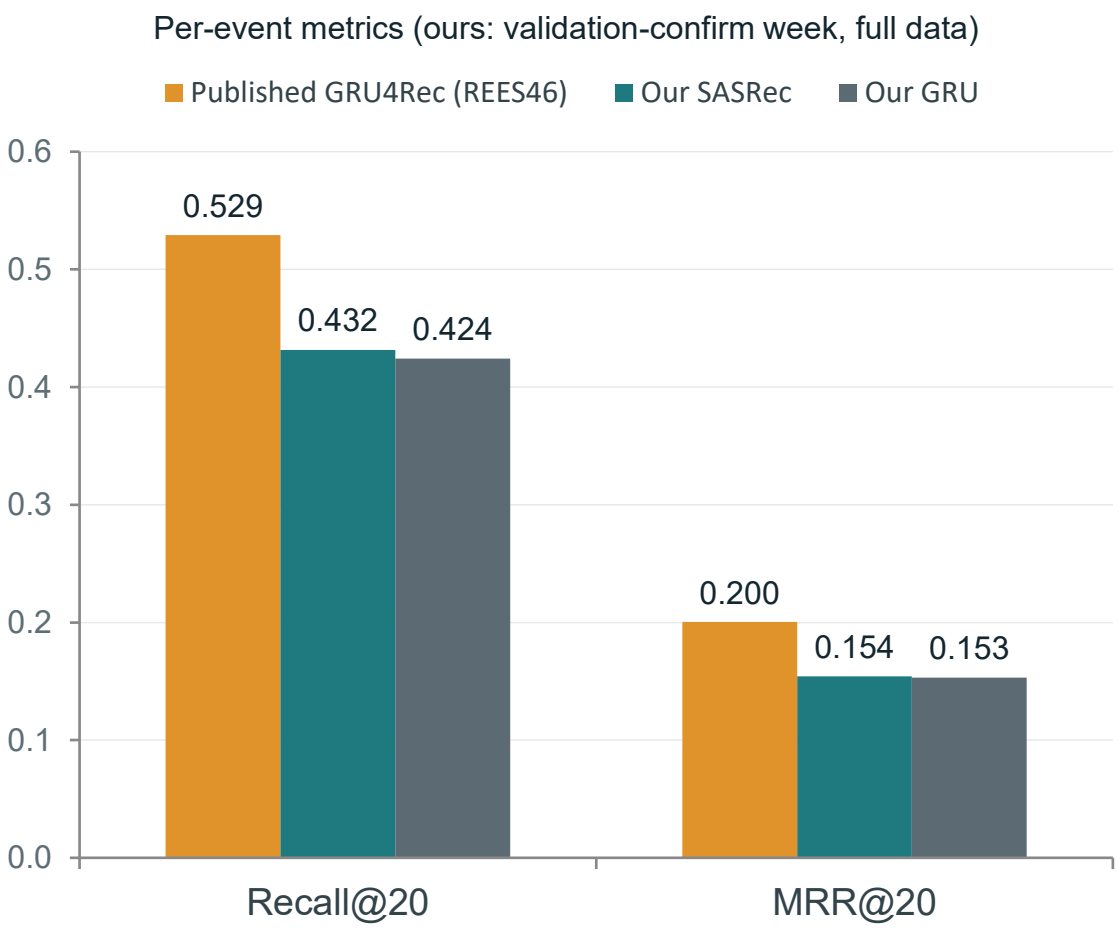
The gain came from the extra feature channels, not the architecture.

Source: Central-model campaign final report §§1, 6.1; planning evidence packet (data census)

OUR NUMBERS VS THE PUBLISHED BENCHMARK

EARLIER PHASE · full data · not comparable with the current study

Below the paper — and why



−0.098 — **Recall@20** — we rank all 158,486 products and count new items as misses

−0.046 — **MRR@20** — we predict the following week; the paper uses a last-day split

5 epochs — **our training** — ≈ 20.7 GPU-h on 2× A16 (15.4 GiB) — the paper is heavily tuned

20× — **less data now** — current study 2.59M vs 52.8M decisions (chosen for federated) → 0.1612

Not a same-protocol re-run. Now: 2× A40 server — experiments running.

Source: Hidasi & Czapp, RecSys 2023 (official GRU4Rec REES46 page) as recorded in the literature register (R05) and benchmark note; central-model report §6.1; planning evidence (A16)

DISCUSSION / DECISION 1

Task and benchmark

1

Do you agree next-product is the right task for the thesis?
Next-category had no published REES46 results and 82% was solvable without learning.

2

Should we re-run the published GRU4Rec protocol on our server?
It would tell us directly whether we are stronger or weaker (≈ 1 extra run, estimate).

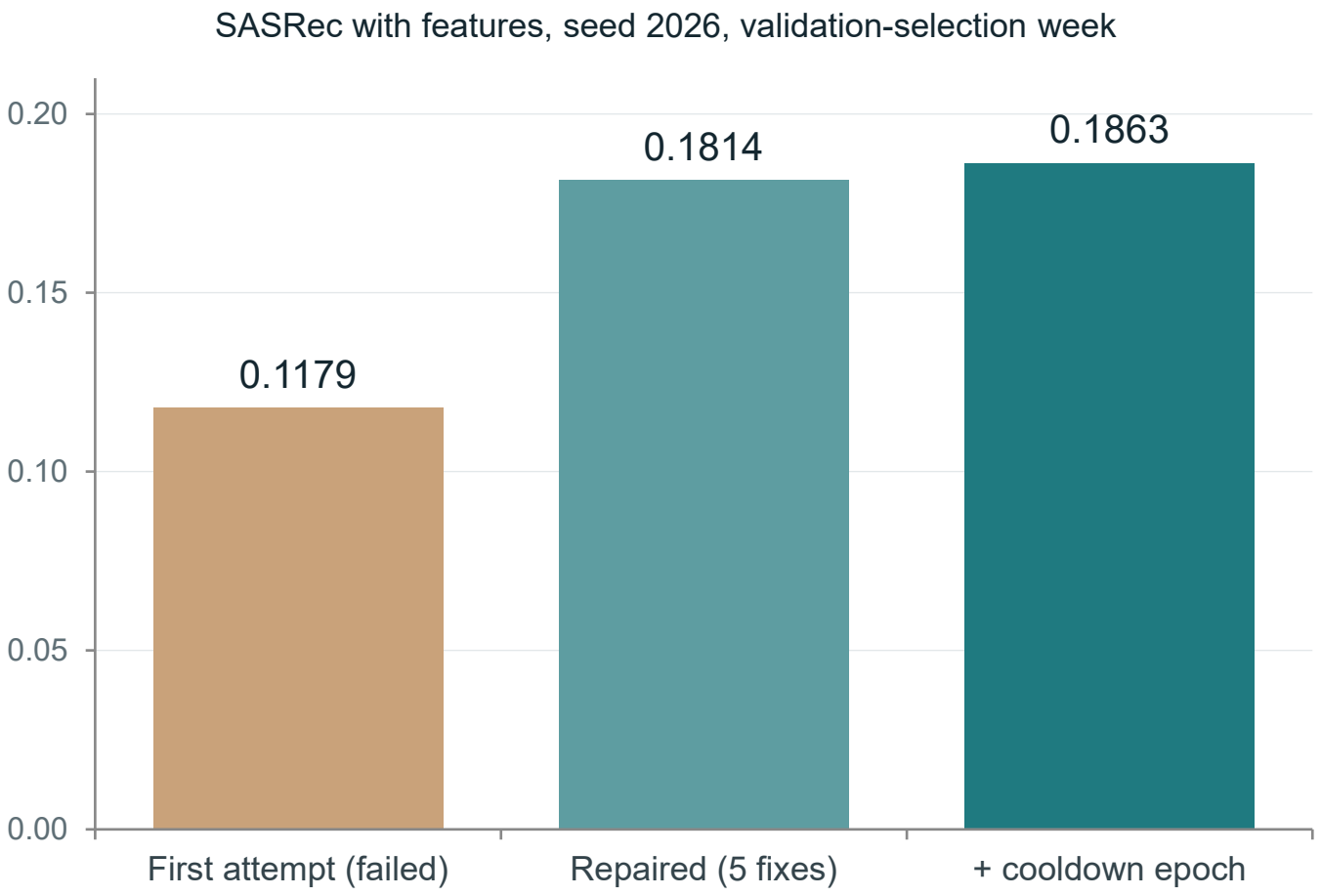
3

Which metric should lead: per-user MRR@20 (ours) or per-event (papers)?
We can report both.

Context: earlier phases vs the published benchmark

EARLIER PHASE · full data · not comparable with the current study

A failed first attempt, fixed and documented



+54%

repaired vs failed attempt (0.1814 vs 0.1179)

Fixes: untied output table · no ×16 input scaling + input layer norm · Adam β2 0.999 · re-centred output table · new learning-rate schedule

0.0002

gap between two seeds at the final epoch (stable)

Source: Central-model campaign final report §§3–4

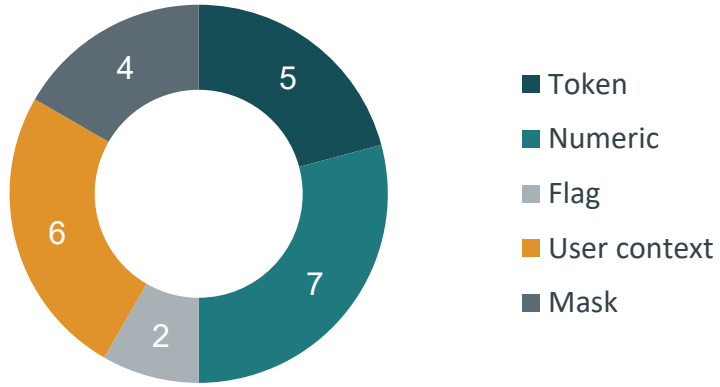
FEATURE ENGINEERING

From 58 input columns to 24

Earlier phase — 58 inputs

6	**token columns**	item · category · brand · main category · daypart · weekday
19	**numeric columns**	price · hour/weekday (sin, cos) · gap since last view · item age · session counts
15	**quality flags**	price missing / zero · brand recovered · category conflict …
12	**user-context values**	past views · sessions · brand / category match share · price vs usual
6	**masks**	history available · previous session available …

Current study — 24 inputs



24/53

kept: statistics filter (→ 30), then grouped search

21 / 24

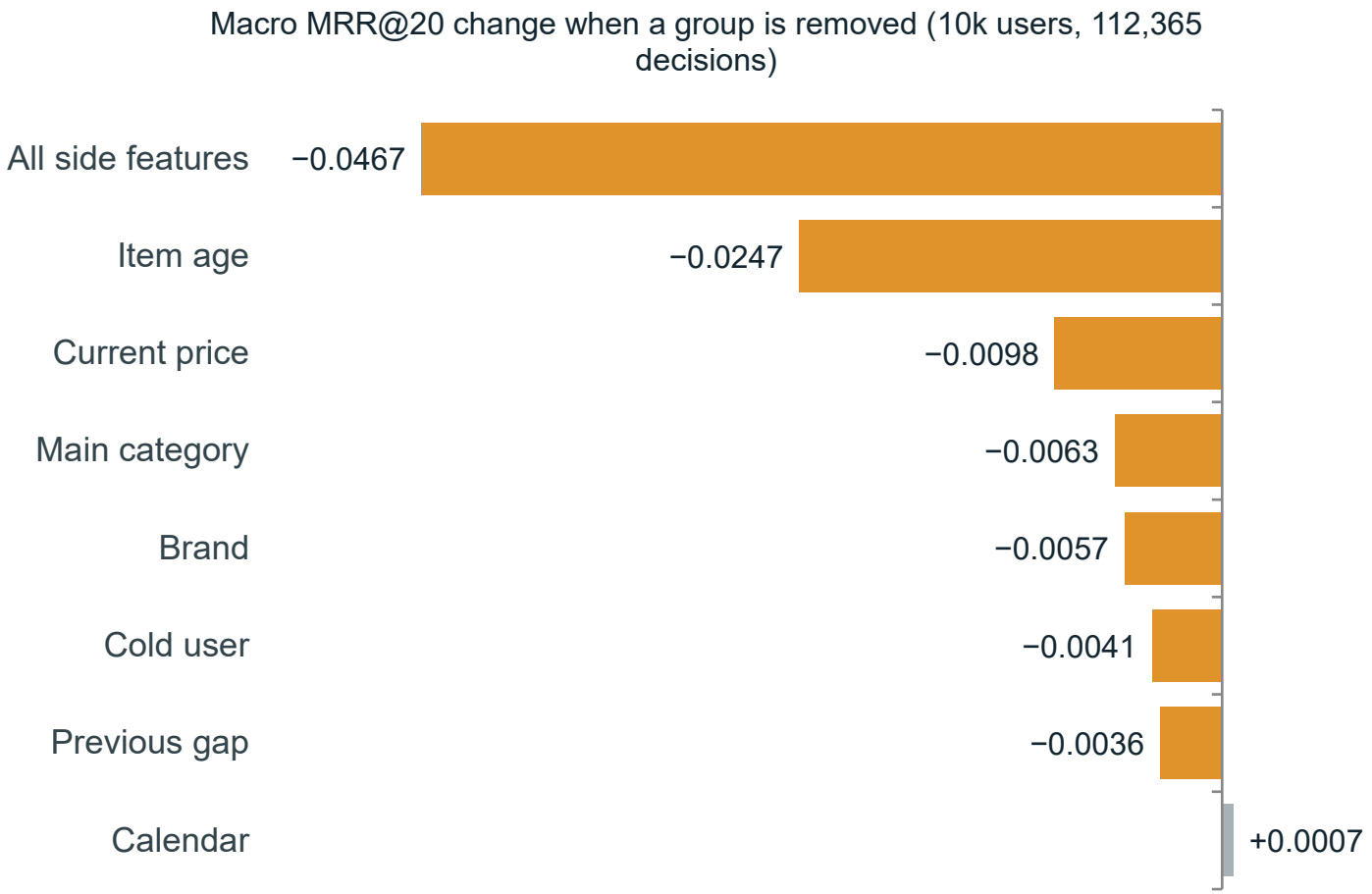
computable on the device

Source: Feature dictionary §15.1; execution reference §2.6; feature search (decision log rows 39, 52–54); review part 4 (features)

FEATURE STUDY

EARLIER PHASE · full data · not comparable with the current study

What each feature group is worth



−0.0467

without any side feature: features matter

+0.00026

best new candidate; threshold 0.002 → keep current

+0.0007

calendar: within noise → dropped later

Source: Feature-study final report §3; living thesis reference (decision KEEP_CURRENT)

WHY WE CHANGED COURSE

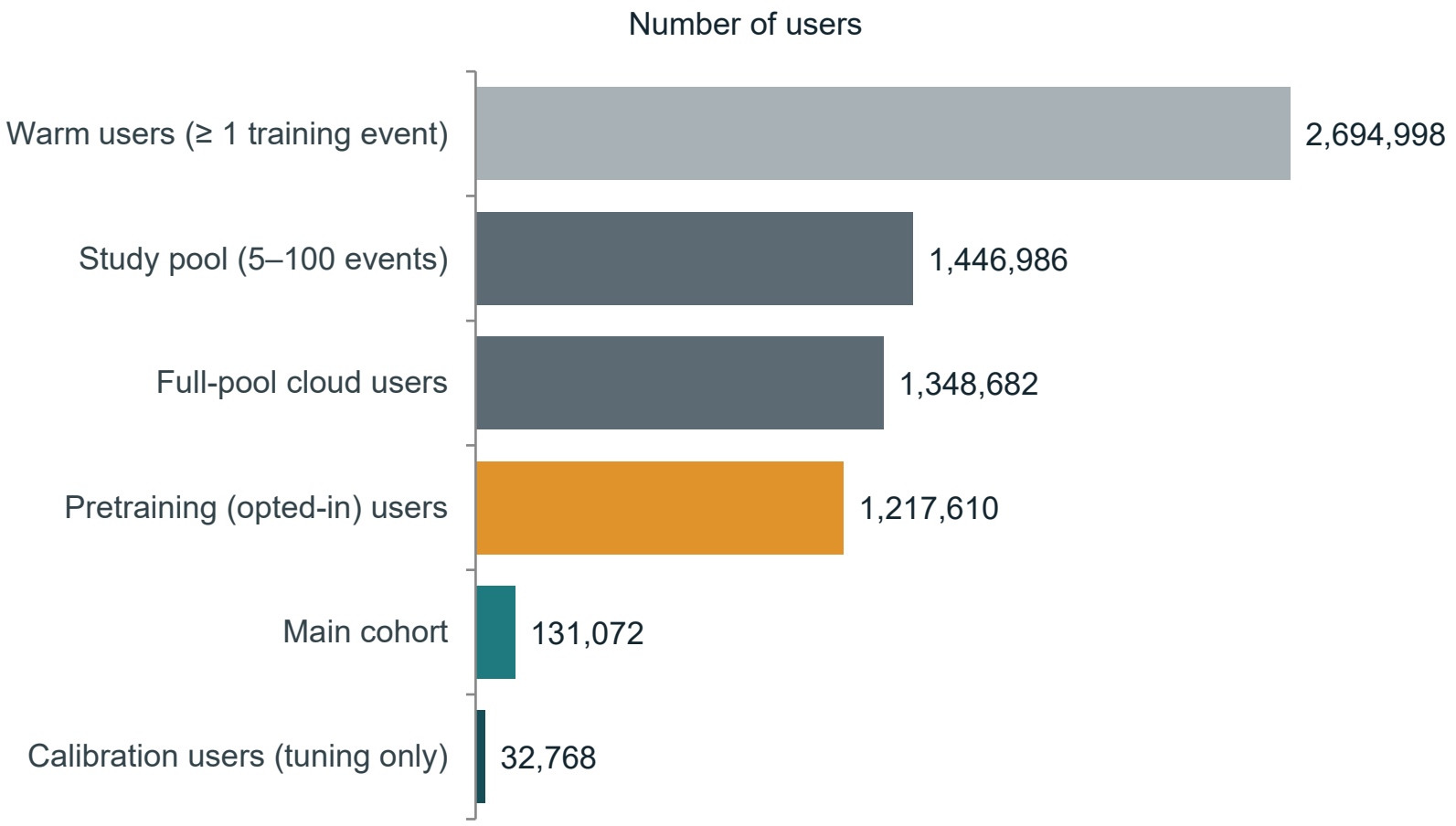
The earlier phases answered a different question

Wrong question	Best cloud model ≠ the proposal's question (cloud vs federated vs on-device)	**→ a deployment study**
One user = one device	Raw users: median 6 events, max 11,628; 473,402 users with a single event	**→ users with 5–100 events: 1.45M**
Federated cost	52.8M decisions is too large to simulate federated training many times	**→ main cohort 2.59M (≈ 20× smaller)**
Unfair family comparison	GRU and SASRec had different features and training schedules	**→ a new contest on one clean protocol**
Test week already seen	The old test week was read once	**→ disclosed as seen; one new sealed look**
Limited compute	2× A16 GPUs (15.4 GiB each), 12.8 GiB free disk	**→ now 2× A40 server; experiments running**

Source: Owner scope for the deployment study; planning evidence packet (data census); population decision; central-model report

THE NEW SETUP

From raw users to the calibration set



Split

Train	1 Oct – 15 Nov
Validation	16–22 Nov
Test (sealed)	23–29 Nov

- Pretraining users never overlap the main cohort.
- Up to 50 past events per prediction.
- Unseen-user panel 12,862; truly cold users 381,295.

Source: Planning evidence packet (data census); population decision; pretraining cohort build; realism audit D2

DISCUSSION / DECISION 2

Is the change of course sound?

1

Is limiting users to 5–100 events acceptable for the Edge story?

It keeps each simulated device realistic; very heavy and single-event users are reported as separate panels.

2

Which cloud should be the reference: 131k matched users or the 1.35M full pool?

We report both — which leads?

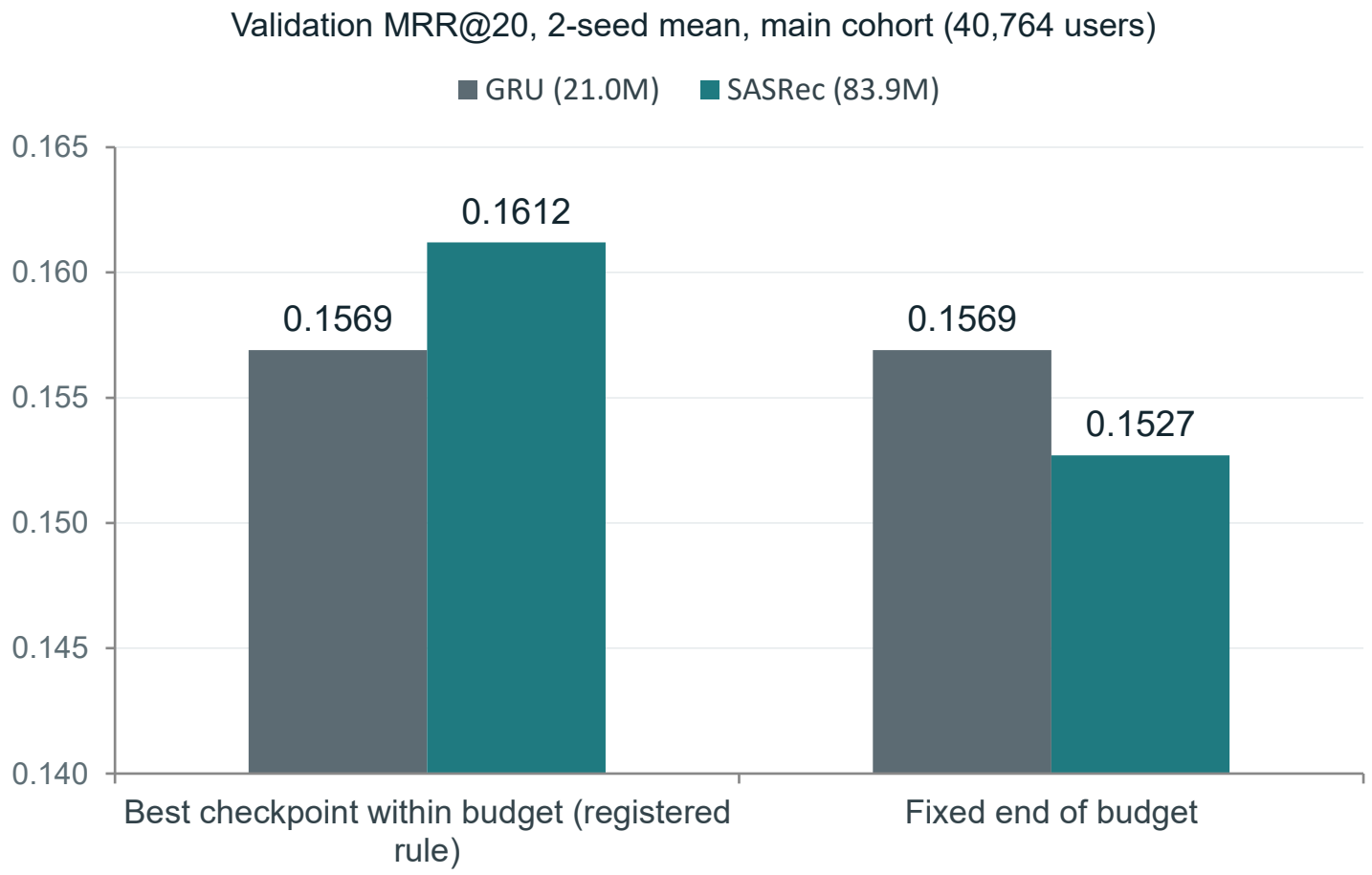
3

Should the earlier full-data results appear in the thesis, or only as context?

They are not comparable (different data size and features).

CURRENT STUDY — BACKBONE CONTEST

SASRec wins under the registered rule; GRU at the fixed end



+0.0043

SASRec, CI [+0.0033, +0.0052] — selected

+0.0042

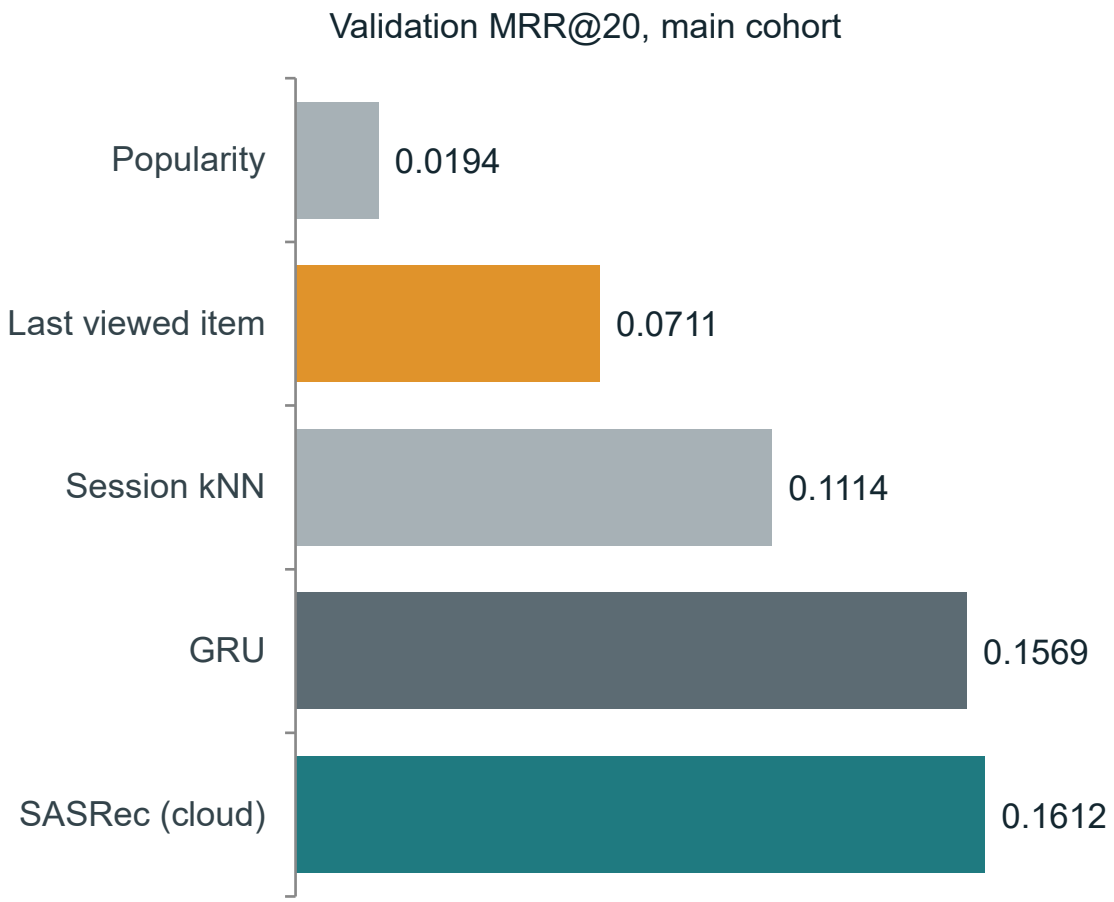
GRU at the fixed end, CI [+0.0033, +0.0052] — disclosed

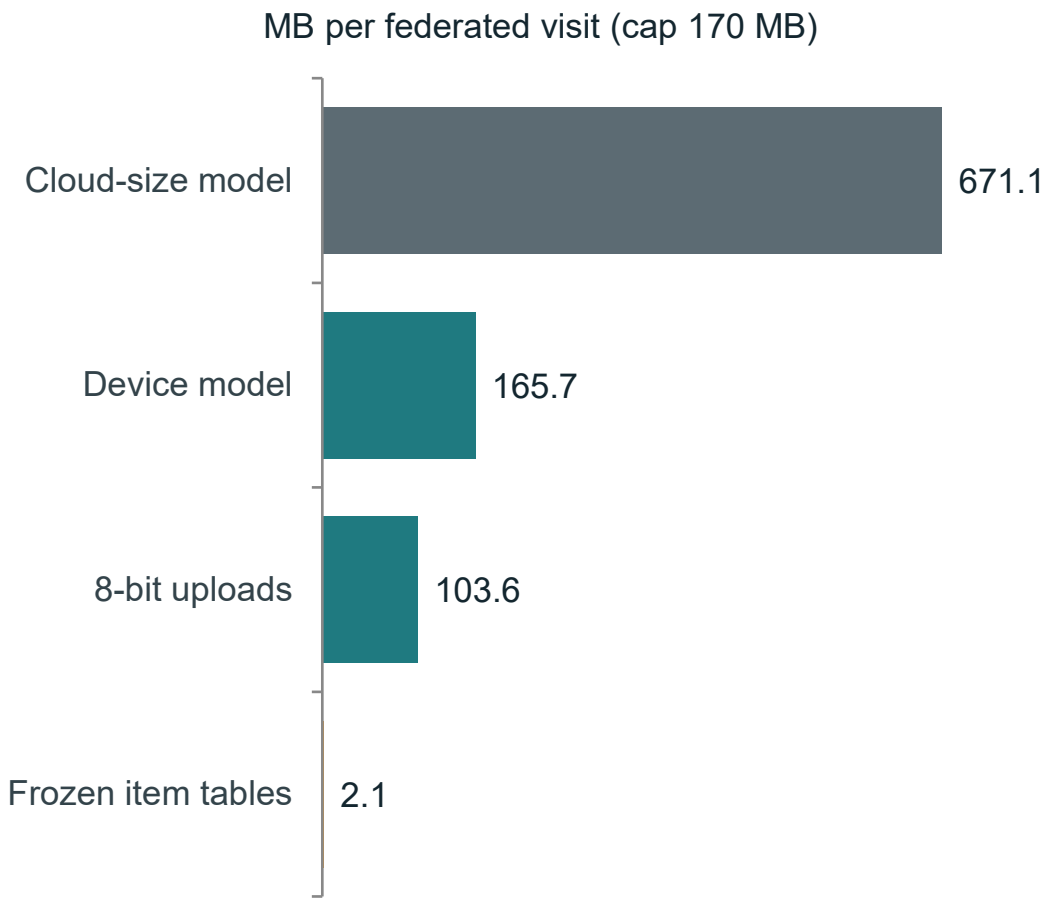
0.1967 vs 0.1910

NDCG@10, SASRec vs GRU

Source: Contest result file; contest diagnostics (decision log, 28 Sep)

Simple rules reach 44–69% of the cloud



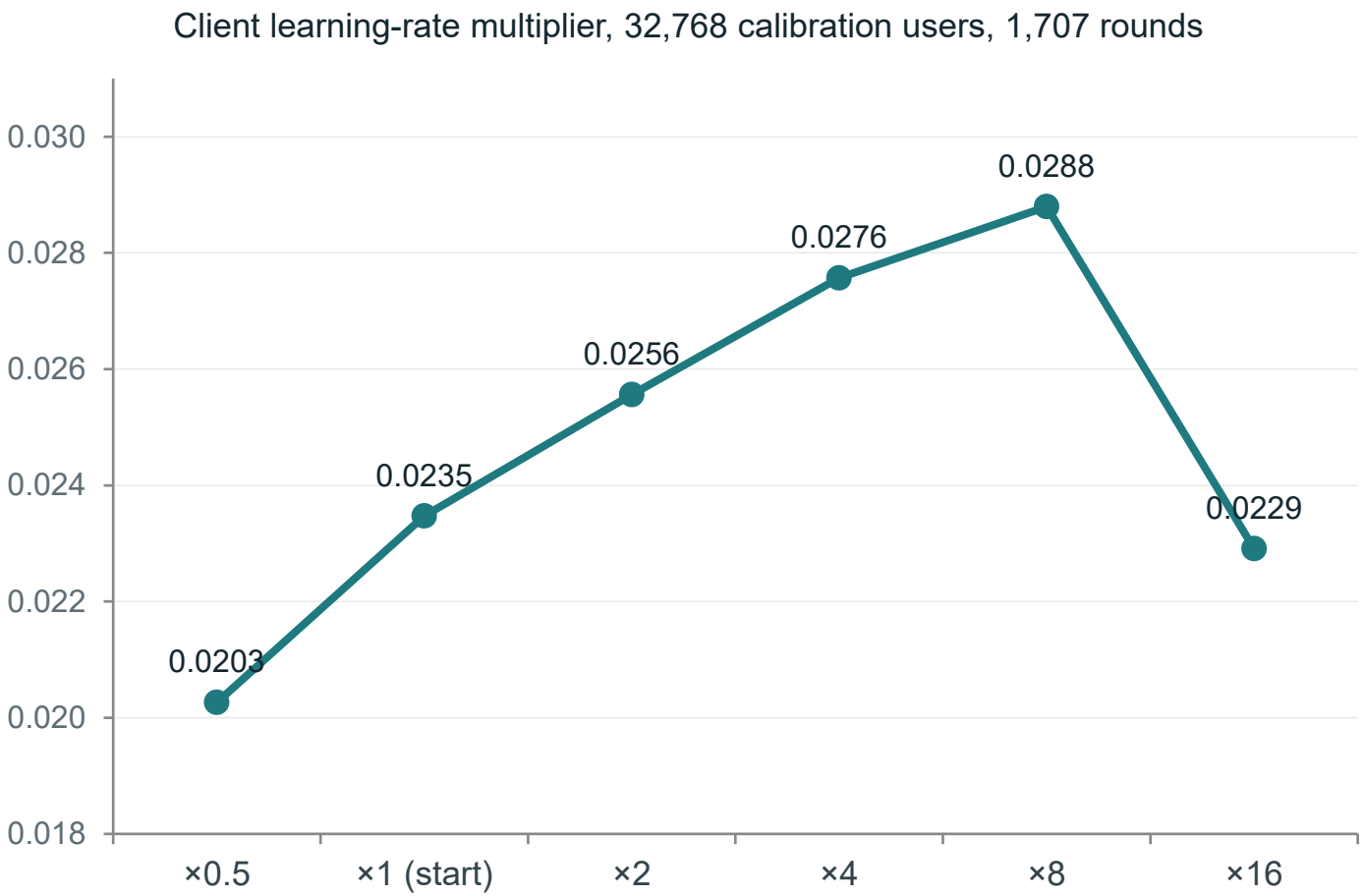


Parameters: 83.9M cloud vs 20.7M device. Frozen tables: ≈ 1 MB each way + one 81.78 MB download (to be re-measured).

Source: Decision log (baselines, 28 Sep); architecture lock (parameters, bytes); review synthesis §3.2

CURRENT STUDY — FEDERATED LEARNING

Tuning so far: 0.0235 → 0.0288 (+22.7%)



≈ 25%

federated vs matched central (calibration, est.)

≈ 44%

“last viewed item” vs cloud (main cohort)

Next stages

Server optimizer	running
LR shape × client optimizer	next
Local passes / batch	next
Pretrained start	running
Frozen item tables	next

Source: Improvement-ladder register; learning-rate selection files; review synthesis §1

DISCUSSION / DECISION 3

Federated effort and privacy level

1

How much tuning effort is fair for federated learning?
Our ladder stops when a stage gains < 0.002.

2

If tuned federated stays below 0.0711 (“last viewed item”), how do we present it?
Plan: report it plainly as a finding about this recipe and budget.

3

Is DP at ε = 8 and ε = 1 enough, or do you want a curve?
ε = 4 costs one more run (≈ half a day, estimate).

Context: federated ladder and baselines

CURRENT STUDY — COMPARISON BOARD

Every method, compared with what, and its status

Level	Method	Compared with	Status
Cloud	Large cloud model	reference	0.1612 (contest)
Cloud	Full-pool cloud (1.35M users)	reference 2	training ends ≈ 30 Sep
Cloud	Small cloud model (device size)	large cloud	sealed
Federated	Federated (tuned), FedProx, personal vector, 8-bit uploads	cloud; floor	tuning stage 2 of 4
Federated + DP	DP at ε = 8 and ε = 1	federated	after tuning
Pretrained branch	Pretrained (1.22M users) → + federated → + frozen tables	full-pool cloud	pretraining running
On-device	Local-only model; cloud / federated / pretrained model fine-tuned on device	cloud; its own base	device runs started
Baselines	Last viewed item 0.0711 · session kNN 0.1114 · popularity 0.0194	floor	done

Sealed results are read together: validation ≈ 2–3 Oct, single test look ≈ 4–5 Oct (estimates).

Source: Contest result; baselines (decision log 28 Sep); run status (decision log 29 Sep); master plan v5

The key numbers

$\varepsilon = 8 \cdot \varepsilon = 1$

client-level DP; noise multipliers 0.5642 · 1.3094

$\delta = 5 \times 10^{-6}$

adding or removing one user's history

1,217,610

opted-in users for pretraining (24.1M decisions)

265,162

parameters uploaded with frozen tables (≈ 1.06 MB each way)

81.78 MB

one-time table download

98%

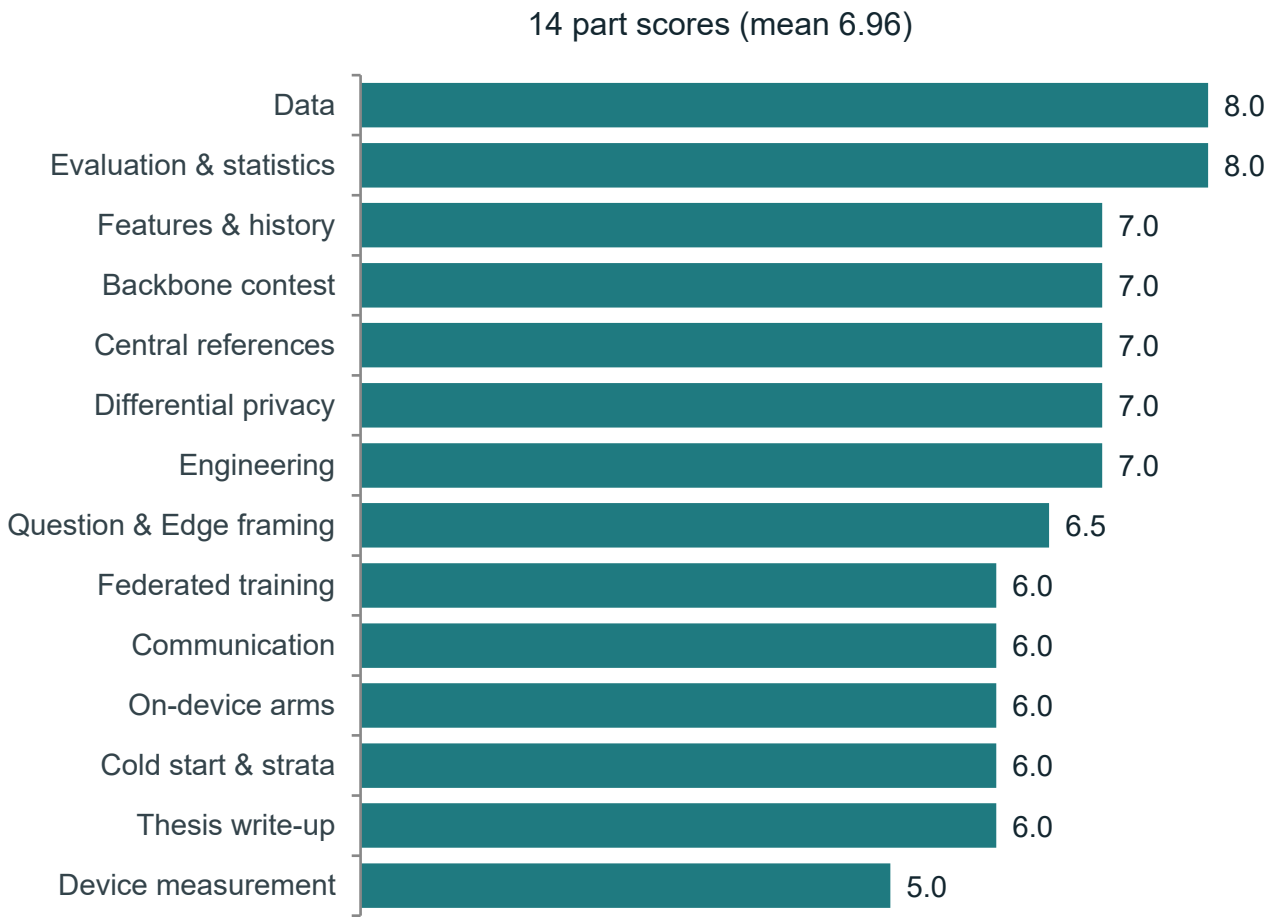
of device-model parameters are item tables → updates can reveal viewed items

Adversary: an honest-but-curious browser vendor. Secure aggregation is not simulated — stated as a limitation.

Source: Privacy statement and threat model; independent DP check; pretraining cohort build; review synthesis §3.2; realism audit

INDEPENDENT REVIEWS

7/10 overall; strongest on data and evaluation



7 → 8–8.5

overall now → after the registered fixes

0

evaluation or data-leak bugs found

3 of 5

fix groups done or registered before any read

Source: Review synthesis (§1 scorecard, §4 fix gates); decision log 29 Sep 17:59–18:36Z

WHEN RESULTS ARRIVE

Four milestones (estimates)

≈ 30 Sep

Tuning and pretraining finish

running

≈ 30 Sep – 2 Oct

Main federated + DP runs (≈ 10.7 h each)

next

≈ 2–3 Oct

One validation read, then lock

sealed

≈ 4–5 Oct

Single test look; tables ≈ 6–7 Oct

sealed

Plan said 1–2 Oct for the first read; ≈ 1 day slip because tuning screens run slower when sharing GPUs.

Source: Master plan v5 (cost model, test-look duration); ladder schedule; decision log 29 Sep

DISCUSSION / DECISION 4

Headline and priorities

1

Which comparison should be the Edge headline?

Plain federated vs cloud, or pretrained branch vs full-pool cloud (the review recommends the latter).

2

If time is short, what do we protect first?

Never cut: federated ×2, local-only ×2, cloud fine-tune ×2, INT8, the lock, the test look.

3

Can you pre-approve the Queen's AI-use disclosure?

Required front matter per the independent review.

Context: cut order is pre-registered in the master plan